

Measuring AI Ability to Complete Long Tasks

Miton AI Times

27.3.2025

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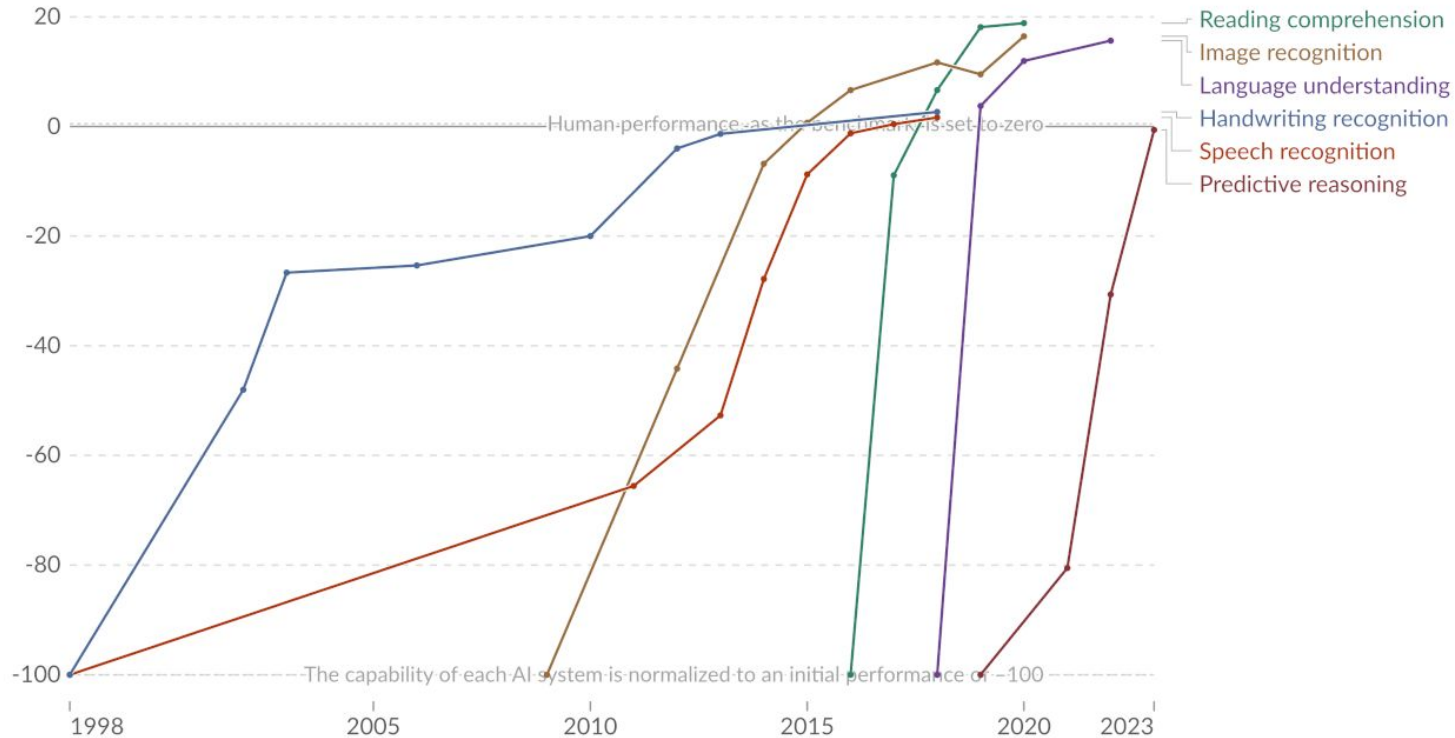
How to measure AI progress

- we have a lot of benchmarks
- most saturate rather quickly
- but what does that mean for real-world use cases?
- hard to say

=> We need something that will not saturate and have interpretable results

Test scores of AI systems on various capabilities relative to human performance

Within each domain, the initial performance of the AI is set to -100. Human performance is used as a baseline, set to zero. When the AI's performance crosses the zero line, it scored more points than humans.



Data source: Kiela et al. (2023)

OurWorldinData.org/artificial-intelligence | CC BY

Note: For each capability, the first year always shows a baseline of -100, even if better performance was recorded later that year.

How to make a better benchmark

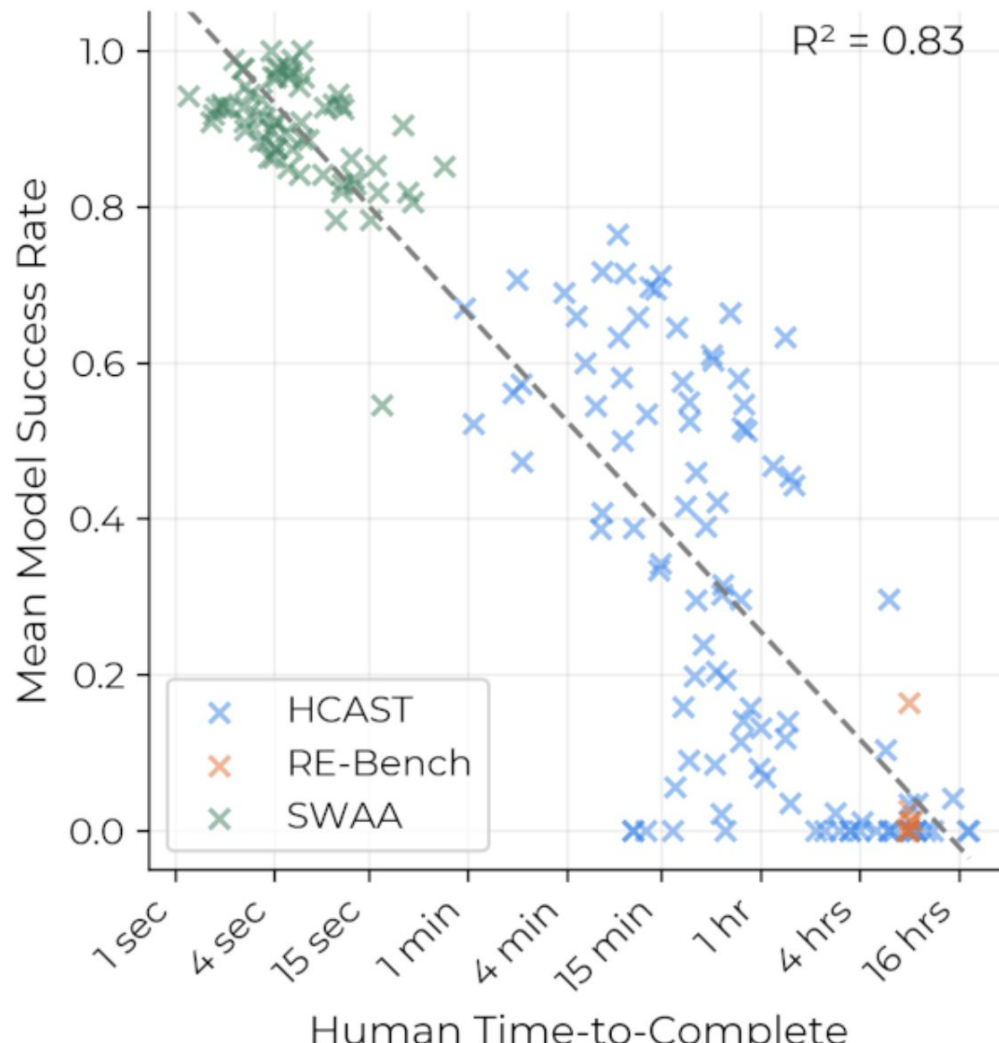
- AI has the individual skills = for **exam-style problems**
- BUT it struggles with completing tasks requiring sequences of actions

=> Let's compare to human performance on tasks not skills

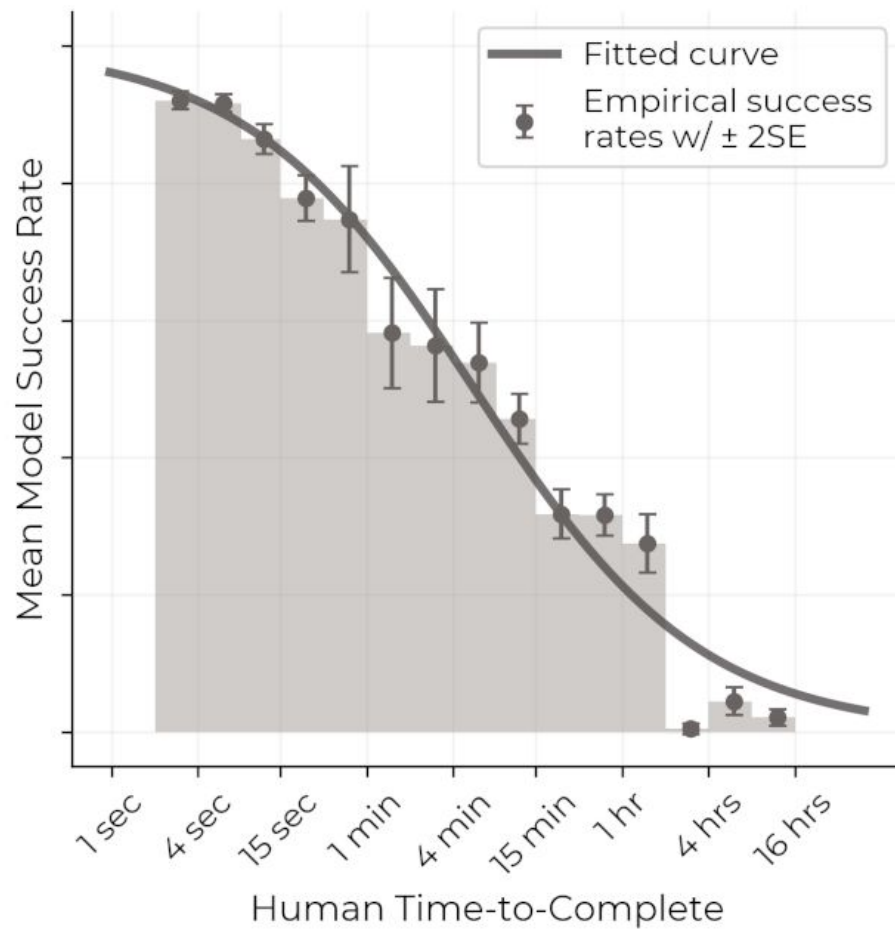
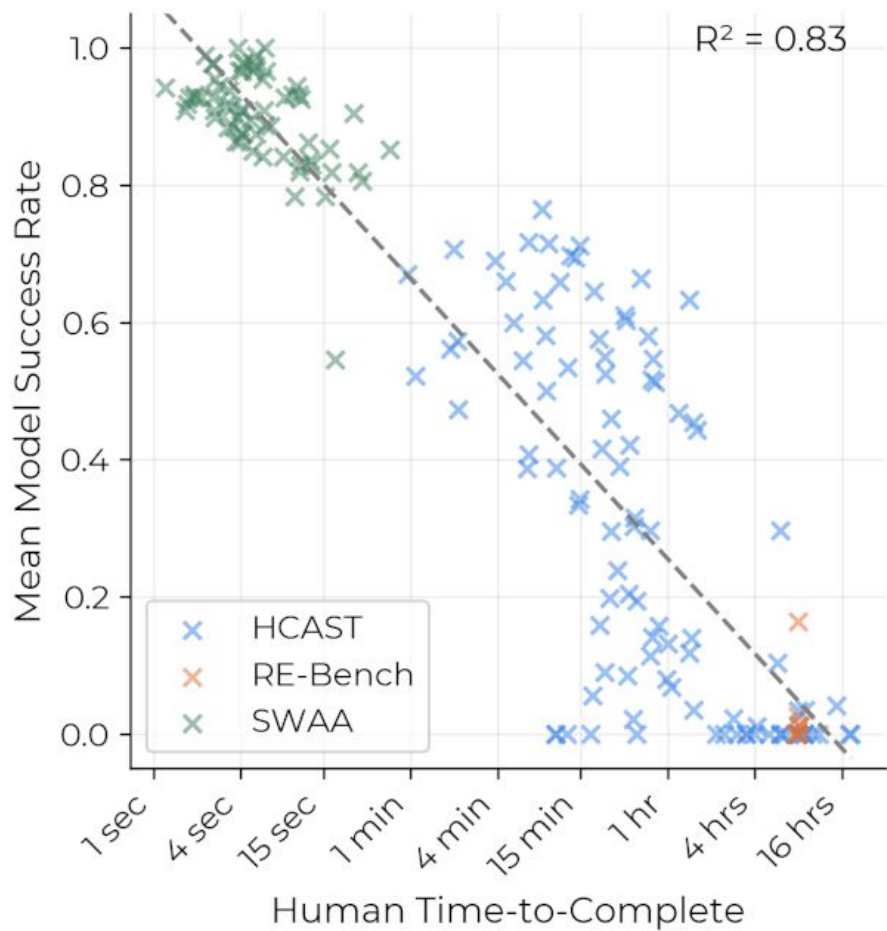
- define diverse tasks
- take humans with relevant experience
- measure the time it takes them to complete the tasks = complexity of task
- see how complex tasks can the AI do?

Length of task predicts AI success rate

- proves it's kinda robust
- different task types have different success rates
- => fit a logistic curve per model over tasks
- pick 50% success rate



Model Success Rate vs Human Completion Time



Proposed benchmark

X%-(task completion) time horizon

- = length of tasks that models can complete approximately X% of the time

Benchmark tasks:

- diverse set of multi-step software and reasoning tasks
- SWAA = 30s programming tasks = easy
- HCAST = minutes - hours long tasks requiring agency = scales with time
- Re-Bench = 8 hours R&D Engineering tasks = impossible now

1 | Diverse Task Suite

HCAST

Diverse tasks that require agency

1 min–30 hrs **97** tasks

SWAA Suite

Single-step tasks sampled from SWE work

1–30 sec **66** tasks

RE-Bench

7 AI R&D Research Engineering tasks

8 hrs **7** tasks

2 | Task Performance



Human Runs



1 hrs



2 hrs



3 hrs



Time Estimate



Agent Runs

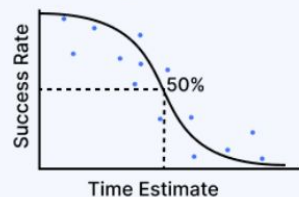


% Success Rate

3 | Time Horizon Analysis

Find Time Horizon

Horizon Length Per Model



Horizon Length



Model Release Dates

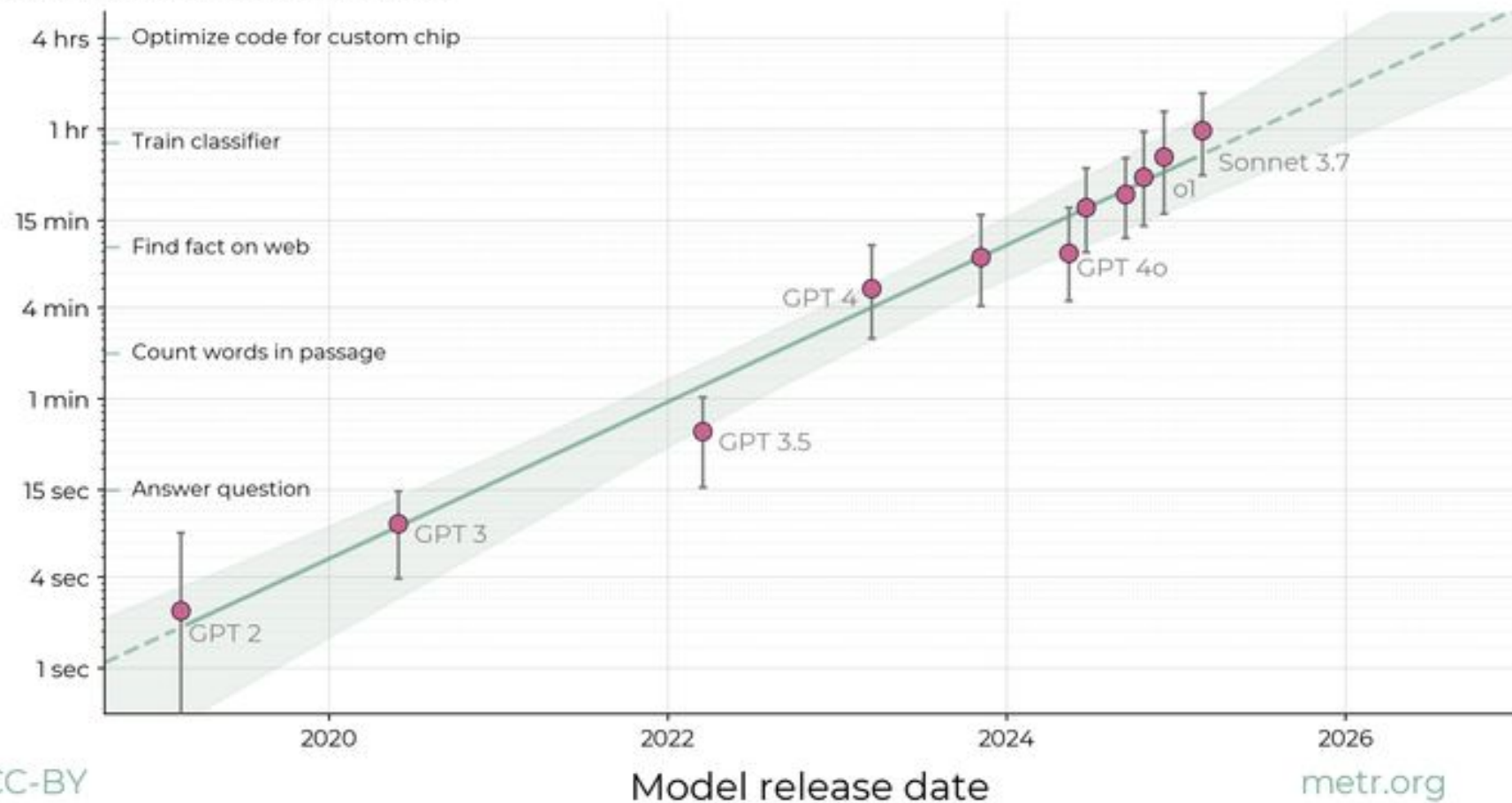
Doubling Time



The length of tasks AIs can do is doubling every 7 months



Task length (at 50% success rate)



Sources

- paper: <https://arxiv.org/abs/2503.14499>
- [Github Repo](#)
- [Summary blog from authors](#)